

JEE MAIN REVISION MASTER: CHEMICAL KINETICS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 34 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Rate Law, Order & Integrated Rate Equations	21 PYQs	Reaction Kinetics
Very High	Arrhenius Equation, Activation Energy & Catalysis	15 PYQs	Temperature & Activation

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Rate Law, Order & Integrated Rate Equations	$\text{First Order: } t = \frac{2.303}{k} \log \frac{[A]_0}{[A]}, \quad t_{1/2} = \frac{0.693}{k}, \quad t_{75\%} = 2 \times t_{50\%}, \quad t_{99.9\%} = 10 \times t_{50\%}$
Arrhenius Equation, Activation Energy & Catalysis	$\log_{10} k = \log_{10} A - \frac{E_a}{2.303 R T}, \quad \log_{10} \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Rate Law, Order & Integrated Rate Equations (Very High Yield — 21 PYQs)

Core Principle: Rate equation: $\text{Rate} = k[A]^x[B]^y$. Units of k : $(\text{mol L}^{-1})^{1-n} \text{s}^{-1}$. Zero order: rate is independent of concentration; $t_{1/2} = ([A]_0 / 2k) \propto [A]_0$. First order: $k = (2.303 / t) \log([A]_0 / [A])$; half-life $t_{1/2} = (0.693 / k)$ (completely independent of initial concentration). General half-life relation: $t_{1/2} \propto [A]_0^{1-n}$.

■ **Caution / JEE Pitfall:** Order of a reaction can be fractional or zero and must be determined experimentally; molecularity is always a positive integer (1, 2, 3) and can never be zero or fractional.

■ **Speed Trick:** In a first-order reaction: Time required for 75% completion is exactly $2 \times t_{1/2}$; for 99.9% completion is $10 \times t_{1/2}$!

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "For a 1st order change $R \rightarrow P$, the concentration of Reactant R changes from 0.1 M to 0.025 M in 40 minutes. The rate of re..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Catalysts are used to increase the rate of a chemical reaction. Because it..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Half-life of a first order reaction is 20 seconds and initial concentration of reactant is 0.2 M. The concentration of reactant left after ..." [Ans: N/A]

3.2 Arrhenius Equation, Activation Energy & Catalysis (Very High Yield — 15 PYQs)

Core Principle: Arrhenius equation: $k = A e^{-E_a/RT}$ implies $\ln k = \ln A - (E_a / RT)$. Graph of $\log k$ versus $1/T$ is a straight line with slope $= -(E_a / 2.303 R)$. Temperature coefficient ≈ 2 (rate doubles for every 10°C rise in temperature). A catalyst provides an alternate reaction pathway with lower activation energy E_a .

■ **Caution / JEE Pitfall:** A catalyst increases the speed of reaction by lowering E_a , but it does NOT alter ΔH , ΔG , or the equilibrium constant K_{eq} !

■ **Speed Trick:** If activation energy $E_a = 0$, the rate constant $k = A$ and the reaction rate is completely independent of temperature.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The activation energy for the reaction $X \rightarrow Y$ is 150kJ mol^{-1} . The change in enthalpy for the above reaction is -135kJ mol^{-1} ." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "For a reaction having three steps, the overall rate constant is $k = \frac{1}{k_1 + k_2 + k_3}$. The values of E_{a1}, E_{a2} and E_{a3} are..." **[Ans: N/A]**
- **[2026 Shift] Q#4 (MCQ):** "Which one of the following graph is not applicable for a 1st order reaction ($R \rightarrow P$)?" **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The activation energy for the reaction $X \rightarrow Y$ is 150kJ mol^{-1} . The change in enthalpy for the above reaction is -135kJ mol^{-1} . Then the activation energy for $Y \rightarrow X$ is

- (A) 280kJ mol^{-1}
- (B) 285kJ mol^{-1}
- (C) 270kJ mol^{-1}
- (D) 15kJ mol^{-1}

Correct Answer: None

Solution Analysis: For the reaction $X \rightarrow Y$ activation energy is $E_a(\text{forward}) = 150\text{kJ mol}^{-1}$ and enthalpy change is $\Delta H = -135\text{kJ mol}^{-1}$. Since ΔH is negative, the reaction is exothermic. This means product Y has lower energy than reactant X by 135kJ mol^{-1} . For the reverse reaction, $Y \rightarrow X$ the activation energy will be the energy needed to go from Y up to the activated complex. Using the relation: $E_a(\text{reverse}) = E_a(\text{forward}) - \Delta H$. Substitute the values: $E_a(\text{reverse}) = 150 - (-135)$
 $E_a(\text{reverse}) = 150 + 135 = 285\text{kJ mol}^{-1}$

Benchmark Problem 2: KCET 2026 (Q#2)

For a 1st order change $R \rightarrow P$, the concentration of Reactant R changes from 0.1 M to 0.025 M in 40 minutes. The rate of reaction when the concentration of R is 0.01 M is _____

- (A) $1.73 \times 10^{-5} \text{ M min}^{-1}$
- (B) $3.47 \times 10^{-4} \text{ M min}^{-1}$
- (C) $3.47 \times 10^{-5} \text{ M min}^{-1}$
- (D) $1.73 \times 10^{-4} \text{ M min}^{-1}$

Correct Answer: None

Solution Analysis: For a first order reaction, $k = (2.303 / t) \log \frac{[R]_0}{[R]}$ Here, $[R]_0 = 0.1 \text{ M}$, $[R] = 0.025 \text{ M}$, $t = 40 \text{ min}$ So, $k = (2.303 / 40) \log(0.1 / 0.025)$ $k = (2.303 / 40) \log 4$ Using $\log 4 = 0.6021$, $k = (2.303 \times 0.6021 / 40)$ $k = 0.0347 \text{ min}^{-1}$ For a first order reaction, the rate is $\text{Rate} = k[R]$ When the concentration of R is 0.01 M, $\text{Rate} = 0.0347 \times 0.01$ $\text{Rate} = 3.47 \times 10^{-4} \text{ M min}^{-1}$ So, the correct option is $3.47 \times 10^{-4} \text{ M min}^{-1}$ Option B.